Daneben! — System Architecture Mind Map

Hochschule Niederrhein (HSNR) · 5. Semester · Mikro-controller · Retrofittable Smart Waste Detector

DANEBEN! SYSTEM CORE

Retrofittable IoT Edge AI & Environmental Deterrent for Waste Bins

1. Dual-Sensing Architecture

External Ground Presence `JSN-SR04T`

- Waterproof ultrasonic transducer (Ø 22mm)
- Monitors pavement zone < 1.5m beside bin
- Low-power polling: 2 pings/sec @ 35mA for 25ms

Internal Bin Fill Level `LID SENSOR`

- Down-facing sealed ultrasonic under bin lid
- Depth < 20cm triggers "Bin Full" status
- Only activates if external dump event occurs

Two-Stage Debounce Filter `GDPR SAFE`

- 8.0-second persistence timer filter
- Pedestrians walking past (< 3s) are discarded
- Zero false alerts; prevents camera snapping faces

2. On-Device Edge AI Vision

Optical Pipeline `OV2640`

- Mounted at 45° downward angle (ground only)
- Captures 96×96 or 160×160 RGB frame
- Direct Memory Access (DMA) to 8MB PSRAM

TinyML Neural Network `EDGE IMPULSE`

- MobileNetV2 / FOMO object detection
- INT8 Post-Training Quantization (~350 KB size)
- ESP-NN vector SIMD acceleration (~220ms inference)

Target Classes `SOFTMAX`

- Empty Pavement / Asphalt Ground
- Plastic Garbage Bags & Cardboard Boxes
- Ignores passing animals, shoes, and shadows

3. Hardware & Power System

Microcontroller ESP32-S3

- Seeed Studio XIAO ESP32-S3 Sense module
- Dual-Core Xtensa LX7 @ 240 MHz + 8MB PSRAM
- Integrated 2.4GHz Wi-Fi + Bluetooth 5.0 LE

Power Subsystem SOLAR MPPT

- 18650 Protected Li-Ion cell (3.7V, 3400mAh)
- 5V–6V 1.5W Monocrystalline solar panel
- CN3791 MPPT Solar charge controller board
- Avg standby current: ~3.4mA (~35 days without sun)

Custom Circuit Board KICAD

- "Inverted T" PCB design matching housing profile
- Status LEDs (D1, D2), tactile button (SW1), JST pins

4. IP67 Enclosure & Mechanics

Housing Geometry POLYCARBONATE

- Dimensions: 120 × 80 × 150 mm (UV-stabilized)
- 20° sloped roof (sheds rain/snow; solar angle)
- Protruding perimeter drip edge over seams

Waterproof Sealing IP67 RATED

- Camera: Ø 20mm Borosilicate glass + UV adhesive
- Speaker: Hydrophobic ePTFE membrane (Gore-Tex)
- Continuous silicone O-ring compressed by M4 screws

Installation & Battery Tube TOOL-FREE

- Bottom threaded battery tube (swap without opening box)
- Rear slots for stainless hose clamps / industrial ties

5. Deterrent & Cloud Telemetry

Acoustic Deterrent I2S MAX98357A

- 3W Class-D amp + 40mm waterproof speaker
- Prompt: "Bitte in die Tonne werfen. Es ist noch Platz!"
- Only sounds if internal sensor reports space

Compliance Logging AUDITING

- Checks if trash is picked up after 5 minutes
- Logs "Warning Ignored" rate with exact timestamps

Fleet Telemetry MQTT / LORA

- Payload: ILLEGAL_DUMPING, BIN_FULL, BATTERY_VOLTS
- Transmits only on state change to save battery

6. Business & Municipal Impact

Target Customers SMART CITIES

- Municipalities & City Cleansing Departments
- Private waste contractors (Remondis, Veolia)
- Housing associations & property managers

Business Model SAAS

- Low-cost hardware lease (~$150/unit)
- Monthly SaaS dashboard fee: $3.00 – $5.00/bin/month
- Dynamic truck dispatch prevents empty routes

Reliability & Lifespan 5 YEARS

- Solid-state (zero moving parts to wear out)
- LiFePO4/Li-Ion cells + solar = 5-year outdoor life